Mixture Models, Hierarchical and Density-Based Clustering

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia Sources: `Lectures/L13b_unsupervised_learning_handbook.pdf` (Chapter 3, pp. 42–60 of the file)

What this file is really about:

$$\text{hard groups} \rightarrow \text{soft membership (EM)} \rightarrow \text{a tree of groups} \rightarrow \text{dense regions of any shape} \rightarrow \text{is a}$$

How to use this file. Mixtures run on seven request latencies, hierarchical clustering runs on the eight training runs, and every number can be checked by hand. Part A builds the methods. Part C is practice with answers.

The second dataset. Seven requests to a deployed model took these latencies, in milliseconds: $10, 12, 14, 22, 31, 33, 36$. They look like two kinds of request, and the request of 22 ms lies between them.

Step-up path

You need. K-means, the normal density and Bayes' rule.

1. Replace hard labels with responsibilities.
2. Alternate E and M steps, and prove the likelihood never falls.
3. Build a whole tree of merges, and compare linkages.
4. Define clusters by density, with core, border and noise points.
5. Test whether any clustering is real, and choose a method.

You can stop when you can run a step of EM by hand and say which method suits a given dataset.

Contents

Part A Maths From the Ground Up

A1 From hard to soft membership

K-means must put every point wholly into one cluster, and a point between two clusters then moves a centroid as if it belonged there completely. A student in a mess hall sitting between two groups of friends belongs to neither completely. A **mixture model** can say 70% window group and 30% door group, and let the student count that way towards each group's average.

A2 The normal distribution and the mixture density

A quantity spread around a centre μ with spread σ is often modelled by the normal density

$$\mathcal{N}(x \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big).$$

Suppose each latency comes from one of two normals, but we never see which. With unknown means, a known $\sigma = 4$, and each component equally likely, the density of a latency is

$$p(x) = \tfrac{1}{2}\mathcal{N}(x \mid \mu_1, \sigma) + \tfrac{1}{2}\mathcal{N}(x \mid \mu_2, \sigma).$$

This is a **Gaussian mixture model**, and the two normals are its **components**.

A3 The E-step: responsibilities

If the means were known, Bayes' rule would give the probability that request i came from component k. The common factors $\frac{1}{2}$ and $1/(\sigma\sqrt{2\pi})$ cancel:

$$r_{ik} = \frac{e^{-(x_i-\mu_k)^2/(2\sigma^2)}}{e^{-(x_i-\mu_1)^2/(2\sigma^2)} + e^{-(x_i-\mu_2)^2/(2\sigma^2)}}.$$

These r_{ik} are **responsibilities**, and for each request they add to 1.

> **Worked example.** Take $\mu_1 = 15$, $\mu_2 = 25$, so $2\sigma^2 = 32$. For 22 ms: $e^{-49/32} = 0.2163$ and $e^{-9/32} = 0.7548$, so $r_{41} = 0.2163/(0.2163 + 0.7548) = 0.223$.

latency	r_{i1} (fast)	r_{i2} (slow)
10	0.998	0.002
12	0.993	0.007
14	0.977	0.023
22	0.223	0.777
31	0.001	0.999
33	0.000	1.000
36	0.000	1.000
sum	**3.192**	**3.808**

A4 The M-step: weighted means

If the responsibilities were known, let request i count towards component k with weight r_{ik}. Differentiating the weighted log-likelihood and setting it to zero gives

$$\mu_k = \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}.$$

The second derivative $-\sum_i r_{ik}/\sigma^2$ is negative, so this is a maximum. When every r_{ik} is 0 or 1, it reduces to an ordinary cluster mean.

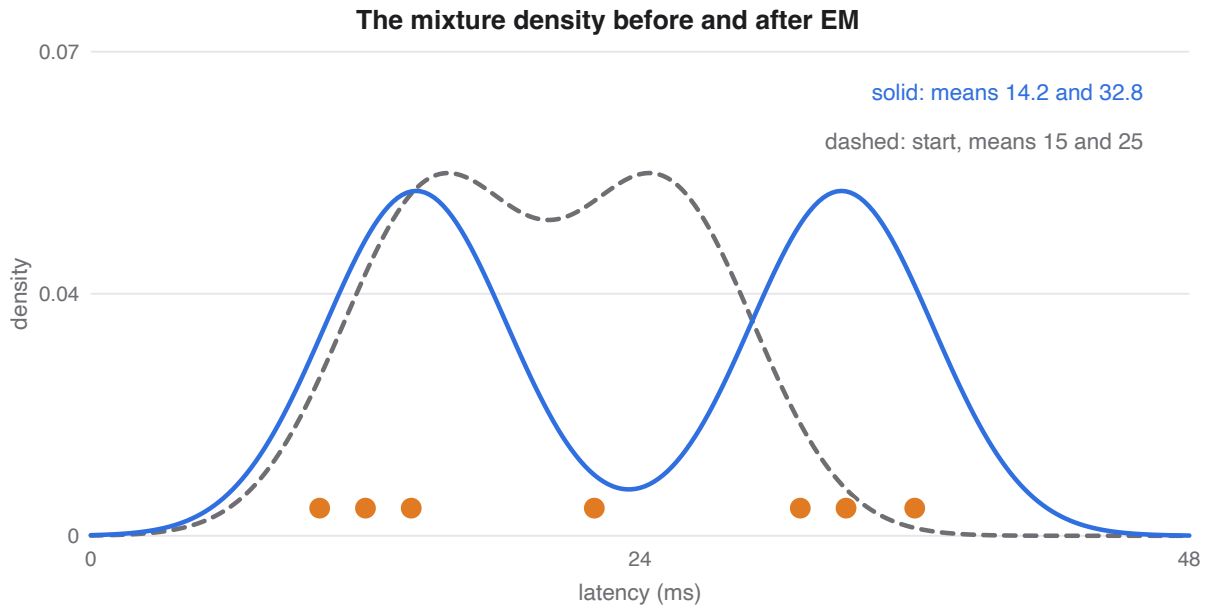
> **Worked example.** $\sum_i r_{i1} x_i = 40.52$, so $\mu_1 = 40.52/3.192 = 12.69$. Since $\sum_i x_i = 158$, $\mu_2 = (158 - 40.52)/3.808 = 30.85$.

A5 The EM algorithm

Neither the labels nor the means are known. Alternate: guess the means, work out the responsibilities (**E-step**), recompute the means (**M-step**), and repeat.

step	μ_1	μ_2	r_{41} for 22 ms	log-likelihood
0	15.00	25.00	0.223	−28.985
1	12.69	30.85	0.436	−24.133
2	13.27	31.54	0.613	−23.980
3	13.70	32.04	0.730	−23.901
4	13.96	32.40	0.795	−23.868

It converges to means of about 14.2 and 32.8.



The mixture density at the start, dashed, and after EM has converged, solid. The dots are the seven latencies. The two components have moved apart to fit the two groups.

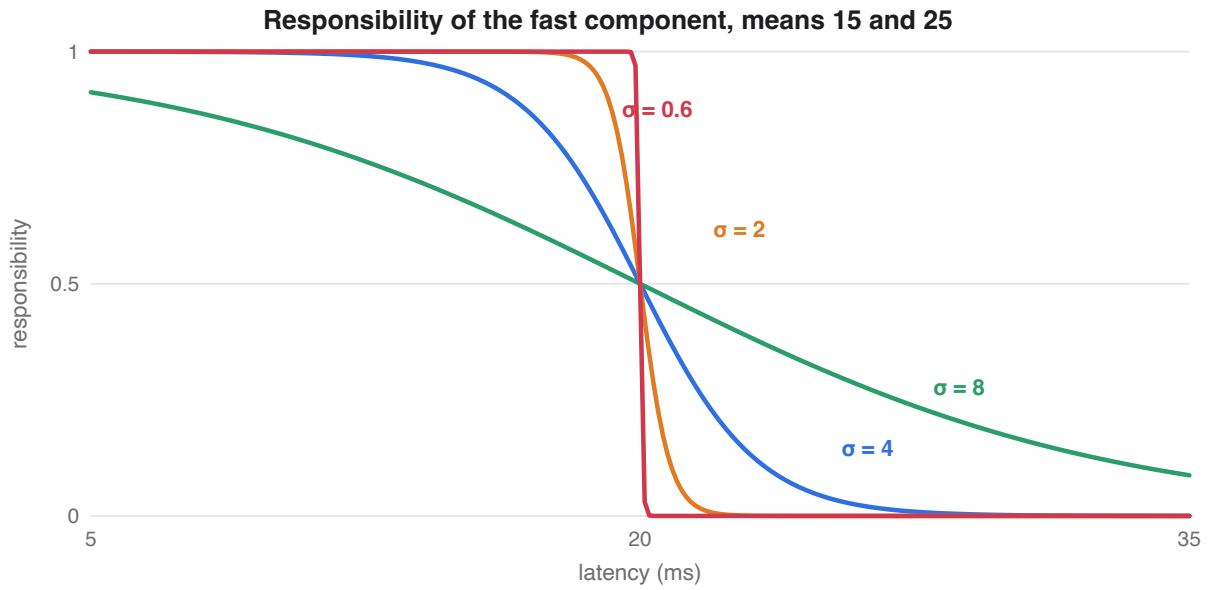
Comparison. K-means from the same start assigns 22 wholly to the slow cluster and stops at 12.0 and 30.5. Neither answer is the truth. Both are local optima of their own criteria, but EM does not force an uncertain point into one cluster, and its responsibilities state the uncertainty.

A6 K-means is the limit of EM

Let μ_a be the nearer mean and μ_b the other, at distances $d_a < d_b$. Dividing through,

$$r_{ia} = \frac{1}{1 + e^{-(d_b^2 - d_a^2)/(2\sigma^2)}}.$$

As $\sigma \to 0$ the exponential vanishes, so $r_{ia} \to 1$: the soft assignment becomes the hard rule of K-means. For 22 ms with means 15 and 25, $d_b^2 - d_a^2 = 40$. So r_{42} is 0.777 at $\sigma = 4$, 0.993 at $\sigma = 2$, and essentially 1 at $\sigma = 1$.



As σ shrinks, the soft curve becomes a step at the midpoint between the two means, which is exactly the K-means assignment rule.

A7 Why the log-likelihood never decreases

Write $p_k(x_i;\theta) = \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k)$ and $\ell(\theta) = \sum_i \log \sum_k p_k(x_i;\theta)$.

Step 1 — A lower bound. The logarithm is concave, so for positive a_k and weights w_k adding to 1, $\log \sum_k w_k a_k \geq \sum_k w_k \log a_k$ (Jensen's inequality). Applying it with any weights q_{ik} gives

$$\ell(\theta) \geq F(q,\theta) = \sum_i \sum_k q_{ik} \log \frac{p_k(x_i;\theta)}{q_{ik}}.$$

Step 2 — The E-step makes it tight. With $q = r$ at the current θ, the ratio inside the log is the same for every k, so the bound equals $\ell(\theta)$.

Step 3 — The M-step raises the bound. It maximises $F(r,\theta')$ over θ'.

Step 4 — Chain them. $\ell(\theta^{\text{new}}) \geq F(r,\theta^{\text{new}}) \geq F(r,\theta^{\text{old}}) = \ell(\theta^{\text{old}})$.

So the log-likelihood never decreases, though it may stop at a stationary point that is not the global maximum.

A8 General Gaussian mixtures

Allow each component its own weight and spread. The E-step is Bayes' rule with π_k, and the M-step is weighted versions of a proportion, a mean and a variance:

$$\pi_k = \frac{1}{n}\sum_i r_{ik}, \qquad \mu_k = \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}, \qquad \sigma_k^2 = \frac{\sum_i r_{ik}(x_i - \mu_k)^2}{\sum_i r_{ik}}.$$

$\pi_k = \frac{1}{n}\sum_i r_{ik}$ comes from a Lagrange multiplier on the constraint $\sum_k \pi_k = 1$.

With several features each component has a covariance Σ_k, and its contours are ellipses:

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\Big(-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big).$$

A small τI is added to each covariance to keep it invertible.

> **Careful.** A component can collapse onto one point with its variance going to zero, and the likelihood then grows without limit. That is why a variance floor or τI is used, and why K-means starts are common.

A9 Agglomerative hierarchical clustering

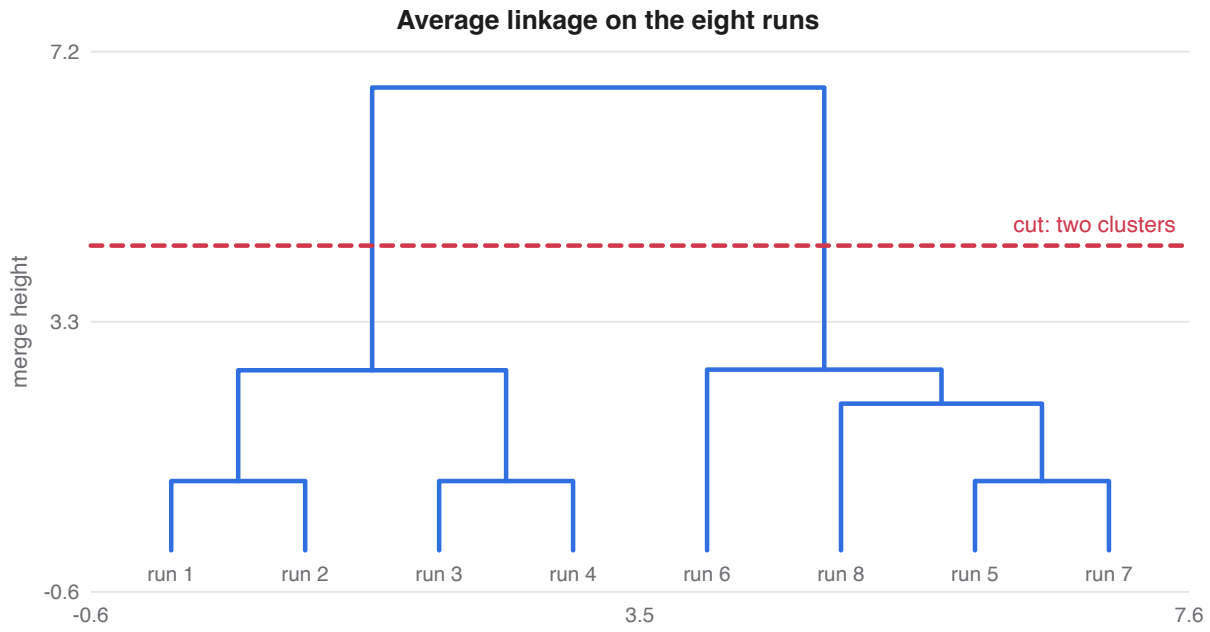
Start with every point alone and repeatedly merge the two closest clusters, recording each merge in a tree called a **dendrogram**. Cluster distance is the **linkage**:

linkage	distance between clusters
single	the smallest pairwise distance
complete	the largest pairwise distance
average	the mean pairwise distance

Worked example, average linkage on the eight runs.

cluster formed	height
1, 2	1.000
3, 4	1.000
5, 7	1.000
5, 7, 8	2.118
1, 2, 3, 4	2.600
5, 6, 7, 8	2.609
all	6.682

Run 8 joins $\{5, 7\}$ at $(2 + 2.236)/2 = 2.118$. Any cut between 2.609 and 6.682 gives the K-means answer, and the big gap between the last two heights signals two well-separated clusters.



Each horizontal segment is a merge at that height. A cut at the dashed line gives two clusters. The large vertical gap beneath it is the sign of well-separated groups.

A10 Ward's linkage

Ward's method merges the pair whose union raises the within-cluster sum of squares W the least. Applying $T = W + B$ to the two clusters alone gives a closed form:

$$\Delta(C_1, C_2) = \frac{n_1 n_2}{n_1 + n_2} \|\mu_1 - \mu_2\|^2.$$

It prefers close means and small clusters. Merging the small and large runs costs $\frac{4\times 4}{8} \times 42.8125 = 85.625$, exactly the between-cluster sum B.

A11 Updating dissimilarities without the data

After merging i and j, the **Lance-Williams update** gives the distance to any other cluster k from the old table:

$$\delta(i \cup j, k) = \alpha_i \delta(i,k) + \alpha_j \delta(j,k) + \beta \delta(i,j) + \gamma |\delta(i,k) - \delta(j,k)|.$$

linkage	α_i	α_j	β	γ
single	1/2	1/2	0	−1/2
complete	1/2	1/2	0	1/2
average	$n_i/(n_i+n_j)$	$n_j/(n_i+n_j)$	0	0
Ward	$(n_i+n_k)/n_{ijk}$	$(n_j+n_k)/n_{ijk}$	$-n_k/n_{ijk}$	0

For single linkage, $\frac{1}{2}(a+b) - \frac{1}{2}|a-b| = \min(a,b)$. Memory is about n^2, and time at least $n^2 \log n$, so the method suits small and medium data.

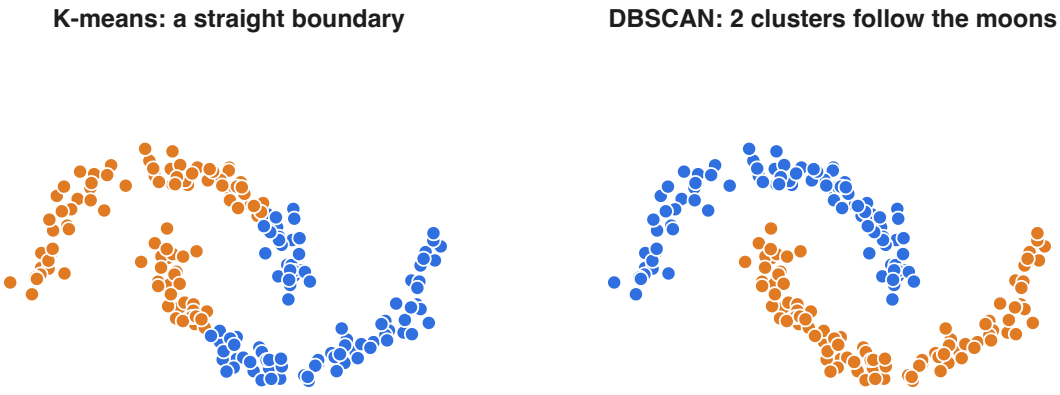
> **Insight.** With two groups joined by a bridge, single linkage chains along the bridge and returns cluster sizes 1 and 94. Complete, average and Ward linkage cut the bridge.

A12 Density-based clustering: DBSCAN

Here a cluster is a region where points are densely packed, separated by sparse regions. Fix a radius ε and an integer minPts.

- The **ε-neighbourhood** of a point holds all points within ε, including itself.
- A **core point** has at least minPts points in its neighbourhood.
- A **cluster** is a maximal set of connected core points, together with the **border points** in their neighbourhoods.
- Everything else is **noise**.

Only core points pass the cluster on, so a border point joins without extending it. The number of clusters is not given in advance, and shapes can be arbitrary.



On two interleaved half-moons K-means splits each moon with a straight boundary, while DBSCAN follows each one as a chain of dense neighbourhoods. Crosses mark noise.

Choosing ε. Because the neighbourhood counts the point itself, a point is core exactly when its distance to its $(\mathrm{minPts}-1)$-th nearest other point is at most ε. Sort those distances in decreasing order and take ε near the bend. In the handbook's example, 200 half-moon points with minPts $=5$: $\varepsilon=0.1$ gives 8 clusters and 74 noise points, $\varepsilon=0.2$ recovers both moons, and $\varepsilon=0.5$ merges them.

A13 HDBSCAN

A single ε fails when clusters have different densities. HDBSCAN defines a **core distance** $\mathrm{core}_k(a)$ as the distance to the k-th nearest other point, and a **mutual reachability distance**

$$d_{\text{mreach}}(a, b) = \max\{\text{core}_k(a),\ \text{core}_k(b),\ d(a, b)\},$$

which inflates distances in sparse regions. It builds a single-linkage tree under this distance and keeps the clusters that persist longest.

A14 Evaluating a clustering

Every method returns clusters, whether or not any exist.

- **Silhouette.** On the 90-point data, the mean is 0.589 for $K = 2$, 0.773 for $K = 3$ and 0.635 for $K = 4$.
- **Comparison with random data.** On 90 uniform points, K-means with $K = 3$ still cuts J by 64% and gives a silhouette of 0.411. The three-cluster data give 93% and 0.773. A moderate score is not evidence of clusters.
- **Stability.** Split the data in half, cluster one half, and compare labels on the other. Agreement is 1.00 on real clusters and 0.84 on uniform data, so judge it against data without clusters.
- **Interpretation.** Examine the mean of each feature within each cluster, and ask whether the clusters are useful for the next task.

A15 Choosing a method

method	a cluster is	K given?	membership	noise label	suits
K-means	points near a centre	yes	hard	no	large data, compact groups
Gaussian mixture	a bell-shaped cloud	yes	soft	no	overlapping groups, uncertain members
hierarchical	a subtree	no	hard	no	small data, several scales
DBSCAN, HDBSCAN	a dense region	no	hard	yes	curved groups, outliers

A common sequence: standardise, reduce with PCA if there are many features, try K-means as a baseline, and move to a mixture or a density method when its assumptions visibly fail.

Part C Practice

P1. With $\mu_1 = 15$, $\mu_2 = 25$, $\sigma = 4$, find the responsibility of the fast component for 14 ms.

$e^{-1/32} = 0.9692$ and $e^{-121/32} = 0.0228$, so $r = 0.9692/0.9920 = 0.977$.

P2. Two points $x = 10$ and $x = 20$ have responsibilities 0.8 and 0.25 for a component. Find its new mean.

$(8 + 5)/(0.8 + 0.25) = 13/1.05 = 12.38$.

P3. Ward's cost for merging clusters of sizes 2 and 3 whose means are at distance 5.

$\frac{2 \times 3}{5} \times 25 = 30$.

P4. Points $0, 1, 2, 10$ with $\varepsilon = 1.5$ and minPts $= 3$. Classify each.

$N(1) = \{0, 1, 2\}$, so 1 is core. 0 and 2 have two points in their neighbourhoods, and are within reach of a core point, so they are border. 10 is noise.

P5. $\mathrm{core}(a) = 2$, $\mathrm{core}(b) = 1$ and $d(a, b) = 1.5$. Find the mutual reachability distance.

$\max(2, 1, 1.5) = 2$.

P6. Why is a high stability score on its own not evidence of real clusters?

Agreement is at least $1/K$ by construction and can be high on structureless data, 0.84 in the handbook's example. It has to be compared with the same score on data with no clusters.

P7. Show that for single linkage the Lance-Williams formula gives $\min(a, b)$.

If $a \leq b$: $\frac{1}{2}(a+b) - \frac{1}{2}(b-a) = a$. If $a > b$: $\frac{1}{2}(a+b) - \frac{1}{2}(a-b) = b$.